



Change is the Only Constant: A Multi-Modal Analysis of Voter Desire for 'Change' in US Elections

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Github: [juanbrasdefer/Modern_Electoral_Truth](https://github.com/juanbrasdefer/Modern_Electoral_Truth)

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List of Abbreviations

ANES - American National Election Studies

APP - American Presidency Project

CCI - Candidate Change Index

CVoter - 'Change-Pattern' Voter: Voter whose voting pattern is consistent with 2012-2020 CCI results

FIPS - Federal Information Processing Standards

MVoter - 'Mixed' Voter: Voter whose voting pattern is inconsistent with 2012-2020 CCI results

SCOTUS - Supreme Court of the United States

TF-IDF - Term Frequency-Inverse Term Frequency

UCSB - University of California Santa Barbara

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Abstract

This paper investigates the role of "change voters" in American presidential elections, distinguishing their behavior from traditional anti-incumbency. Using campaign documents and debate transcripts from the American Presidency Project, we develop a text-based measure of candidate change rhetoric and construct a Candidate Change Index (CCI). We then use American National Election Studies (ANES) data to identify voter patterns consistent with change-seeking behavior and theorize key attributes of pure change voters. Finally, we validate this theory by mapping continuous measures of voter change propensity to candidate CCI scores. The findings suggest that change voting reflects persistent dissatisfaction and instability rather than simple opposition to incumbents, but also highlight key limitations of our data and approach

Introduction

1.1 Introduction

In their 1973 song, 'Money' Pink Floyd sing *"Money, Get Away, Get a good job with more pay and you'll be okay"*. But the American dream being sold in the song has vanished. The stagnation in real wages in the last twenty-five years (Machin 2016) has created an environment of persistent economic distress for many, leading to a subconscious search for change.

Similarly, The Clash tell us in their 1982 hit "Straight to Hell" that *"It could be anywhere, most likely could be any frontier, Any hemisphere, No man's land, There ain't no asylum here"*, painting a picture of the increasingly global reach of wars and the devastation they bring. The increasingly globalized world and the reach of news and negative media have only contributed to this persistent stress, furthering the individual's desire for a new state of affairs.

But how does the individual effect change in their world? Though it may be an un-glamorous answer, the primary way that most people get to try to change the system under which they live is through elections-- imperfect though it may be. The question becomes, then, how does the distressed voter, who seeks the intangible concept of 'change', vote?

1.2 Research Question and Motivations

This is an important question to answer, as an individual voting for 'change':

- 1) Arguably represents a portion of the voting pool type of voter that is likely to grow as the economic and world situations continue to worsen.
- 2) Prioritizes 'change' over partisanship, and thus exists in the prized pool of swing votes.
- 3) Is at least partially voting from a place of subconscious distress, meaning that their failure to recognize the true distressing factors in their entirety makes them more susceptible to external manipulation and rhetoric, potentially by demagogues and populists.

Research Question: How can we conceptualize of a voter who is motivated by the idea of 'change' above all else, and can we use this formulation to identify their propensity for voting for certain presidential candidates?

Literature Review and Contribution

2.1 Contemporary Electoral Motivators

The existing literature on this subject is exceedingly vast.

The consensus on traditional predictors of voter preferences that continue to be strongly relevant today are a mix of economic indicators, such as unemployment (Part et al, 2020) and income level (Gelman et al, 2010), and diverse racial indicators such as ethnicity (Philpot, 2018) and status in the country (Mayda et al, 2022). Authors also agree that certain predictors have risen massively in recent elections, particularly since 2016, including education level (Ahearn et al, 2023) and gender (Philpot, 2018). Education levels are largely seen as existing on strong partisan lines, with higher-education individuals voting overwhelmingly for the democratic party, while lower education levels are associated with republican voters. Similarly, gender follows a striking pattern of partisanship, with women tending strongly towards democrats and men towards republicans, especially in the younger age categories (Philpot, 2018). Crucially, there exists a gap in studying certain latent traits in individuals, specifically something as intangible as a desire for 'change'.

2.2 Candidate 'Change' Posturing and Voter Perceptions

Candidate posturing is similarly a topic that enjoys great study in the literature, ranging from psychoanalysis to marketing. The specific literature relating to candidates that embody 'change' agrees that traits such as honesty, competence'(Lyons, 1997), and being a political outsider (McGraw, 1991) are crucial. More recently, it has become clear that candidates who succeed in credibly sell their 'outsider' status enjoy an added boost, as it delivers a kind of 'anti-incumbency' effect against the political establishment as a whole. (Barr, 2009). Perhaps more importantly for the purposes of this paper, authors identify trends of anti-incumbency rhetoric more strongly in

periods of economic downturn (Abramson et al., 1995). We see then that the posturing of candidates as candidates that can bring 'change' is by no means a new phenomenon, but that it does increase in times of economic strife.

2.3 Voting for 'Change'

The idea of voting for a candidate on the idea of seeking 'change' (conceived of as something related to but distinct from anti-incumbency) is relatively understudied in the literature. Levine et. al (2001) inspect the way that young voters interpret the messages of 'change', "both in an overall sense and as attributed to each of the candidates" (Levine et. al, 2001). The authors find different types of 'change' interpretations by respondents, as well as differing levels of relevance for different demographics. Wüst (2022) studies the German landscape, and similarly finds a strong effect in the youth, seeing that they were the predominant vote in the new political parties (FPD, Greens) and away from the classic coalition of SPD/CDU. Overall, the greatest predictor of change seems to be age, which is unfortunate given the data that this paper has access to. Still, there is a large gap in the study of the change-voting phenomenon, which this paper will contribute to closing.

2.4 Contribution

This paper contributes to the existing literature in formulating a Theory of Change Voters that is empirically founded in candidate posturing, external perceptions of candidates, and rigorous statistical testing to arrive at a phenotype of a change voter that is agnostic of partisan divides and more robust than simple explanations of anti-incumbency.

Data

3.1 American Presidency Project (Campaign Text Data)

Data on candidate campaign text documents (speeches, interviews, press releases...) is taken from the University of California Santa Barbara's (UCSB) 'American Presidency Project' (APP). The APP is functionally a database housing an impressively comprehensive multi-media collection of documents relating to American presidential contests. For the needs of this study, two collections were considered for the basis of a textual analysis: A) Debate Transcripts from Primary and Presidential Debates, and B) 'Campaign Documents', which include Transcripts from Rallies, Interviews, Press Releases, and Statements.

The Debate collection was discarded for two reasons. First, the conclusion was reached that the content and format of a debate is not as informative a resource for deriving candidate posturing, largely due to the fact that the format encourages improvisational response over carefully worded discussion. Second, the small number of debates that take place in each electoral cycle and the short length of these debates creates a problem for our kind of quantitative analysis; there is simply too little data for robust textual analysis.

For this reason, the collection of 'Campaign Documents' was instead to help determine Candidate 'Change' Posturing. This data was scraped from the APP's website using the software R (given a lack of available Application Programming Interface [API]), resulting in a corpus of 21,512 documents¹. This corpus spans the seven electoral cycles (2000, 2004, 2008, 2012, 2016, 2020, 2024) we wish to study, and includes seventy-seven unique speakers², which are made up primary and presidential candidates.

3.2 American National Election Studies (Voter Survey Data)

Voter data is taken from the American National Election Studies (ANES) survey, which is a cross-sectional³, federally conducted survey of the American population aimed at capturing the

¹ Questions of document availability per candidate and the potential relationship between degree of change rhetoric and duration of candidature are covered in the Appendices.

² 'Speaker' indicates the primary speaker in the document, and is a tag taken directly from the APP UCSB data, not inferred or devised.

³ Panel data does technically exist for 2016-2020, but this combination of years alone does not help our study, as it does not increase the number of vote-recall questions available to us. This is further discussed in section 6.1.

social and political beliefs of the population. The survey is conducted every two years, coinciding with general and midterm electoral cycles (from which we take the surveys of 2000, 2004, 2008, 2012, 2016, 2020, and 2024) and consists of (on average) ~1700 total questions. The number of people surveyed also varies across years, with most sitting around 3,000 respondents and a minimum of 603 respondents (2004).

Several challenges exist in the data. First, and crucially, there is an incredible amount of variation in the questions asked in each year's survey, which complicates our analytical attempt to generalize across election years. While core themes (such as economic situation, perception of politicians...) tend to be present across all surveys, the data is admittedly sub-optimal in this regard. Similarly, the ANES does not share any demographic variables of respondents (age, gender, race), which greatly restricts the depth of insights that can be drawn from the results of this paper's experiments. Further, the data is cross-sectional (non-panel), which limits the voting pattern complexity that we are able to examine. As well, Finally, at the time of this paper's writing, the post-election survey results for 2024 have not been published by the APP, meaning that this year unfortunately falls out of scope for our second study.

Study One: Identifying Change Candidates

The first study of this paper profiles the level of 'change' posturing that each candidate attempts as well as the candidate attributes that contribute to their perceived level of 'change'. To create these profiles, two approaches are used, which build on each other.

Methods

4.1 Textual Analysis of Campaign Rhetoric

The first method for candidate profiling is a textual analysis on candidate vocabulary. The idea here is to look at the words that candidates use in their speeches, interviews, and press releases to determine the level of use of 'change' rhetoric that each candidate employs. The basis for this textual analysis is the corpus of 'Campaign Documents' scraped from the UCSB's APP, which is then grouped for analysis by candidate and election year.

The text analysis itself employs a relatively simple dictionary-based frequency count approach, which takes a set of words relating to the topic of 'change' and counts how often they each appear in each document. The dictionary (set of words) used in this study can be found in the Appendices, and was built for this paper around four aspects of 'change': "Direct Synonyms", "Contrast to Status Quo", "Forward-Looking", and "Action Oriented". For each of these aspects, specific verbs and nouns were identified⁴ and stemmed⁵, resulting in a dictionary of fifty-seven stems. Importantly, the dictionary only employs unigrams⁶, with the goal of reducing the often-messy endeavour of matching multi-word grams.

The results of this dictionary-based frequency matching are tallied across documents and years to arrive at a 'percentage use of changegrams' for each candidate.

Importantly, no 'stopword' or punctuation removal was carried out on the text for two reasons. First, this paper is not interested in the proportion of changegrams used by candidates *per se*, but is instead concerned with the *relative* use across candidates, which is agnostic to absolute usage, meaning that 'untethered' percentages are perfectly valid for our use. Second, we are focused only on unigram stems, which inherently have no stopwords or punctuation marks.

4.2 Static Attribute Scoring and Composite Index Construction

Following the results the textual analysis, we create a more comprehensive measure of how candidates embody change. To do this, we create a Composite Index (CI), which helps with tackling concepts that are not easily measurable by combining variables related to the unmeasurable concept and 'approximating' its true (unmeasurable) value. For our study, we conceptualize a candidate's 'degree of change' as being made up of two halves: a candidate's 'Posturing' (the image they put forth) and their 'Perception' by voters (attributes and traits outside of the immediate control of the candidates)⁷.

⁴ Word-Embedding was considered for this task and is discussed in section 9.2: Future Work.

⁵ 'Stemming' in the context of Text Analysis typically means shortening a word to a basic character stem. For example, "innovation" and "innovate" are both shortened to become the same stem, "innovat".

⁶ Unigrams being single-word objects ("forward", "establishment"), as opposed to ngrams (for any number 'N') such as bigrams ("looking forward").

⁷ Importantly, we exclude policy proposals/ platforms from this measure, as our study believes that it is low-information voters who are voting for change, and thus are less influenced by policy (in comparison to Posturing and Perception).

The 'Posturing' aspect of a candidate is derived from the results of our textual analysis of campaign rhetoric, and modifies the aforementioned 'Percentage use of Change Grams' score to become a 'Relative', 'In-Year' measure as follows:

For each Electoral Cycle:

- Consider Only Candidates in the Presidential General Race (ie: Post-Primaries):

- Identify the Top In-Year Change Speaker by '*% Use of Changegrams*'

- Set this Speaker's '*% Use of Changegrams*' Value as the '*Max in Year*'

- For each Candidate in this Contest (typically only two Candidates):

- Divide the Candidate's '*% Use of Changegrams*' by the '*Max in Year*' to arrive at the Candidate's '*Relative in-Year Changegrams*'

The 'Perception' aspect of a candidate is built as a combination of six 'static' or 'semi-static' personal traits, being:

A) Visible Minority - Coded as 0 if the candidate is of 'white' descent, 1 otherwise.

B) Gender - Coded as 0 if the candidate is male, 1 otherwise.

C) Political Dynasty - Coded as 0 if anyone in the candidate's immediate family (ie: sharing a surname) has previously held the office of the President.

D) High Governmental Office - Coded as 0 if the candidate has been President before, 0.2 if the candidate has held a presidential cabinet position (Vice President, Secretary of State...), 0.4 if the candidate has been a Governor or Senator, 1 otherwise.

E) Party Affiliation Relative to Incumbent - Coded as 0 if the candidate belongs to the same party as the incumbent President, 1 otherwise.

F) Age - Coded as the result of taking the candidate's age at the time of the election⁸ and passing it through the following transformation:

$$\text{Age Score} = -1(\log_{10}((\text{Age} - 35) / 100))$$

⁸ Age on September 1 in the election year, when campaigning is in full swing.

Which subtracts the constitutional minimum age for presidential candidacy (35) from the candidate's age, divides the result by 100, and takes the Logarithm of the result. This value is then subtracted from 1 to give our score the correct 'direction' (a younger candidate means *more* change, but their age is a lower number, so the 'direction' has to be changed to give lower ages higher scores), resulting in a value falling between [0,1].⁹

All of these variables are then taken together to construct our Candidate Change Index, bound between 0 and 1, where a value of 1 indicates a candidate that perfectly embodies change.

Weighting of variables is set as:

Posturing:	0.5	
Perception:	0.5	
Of which its sub-components receive:		
Minority:	0.83	
Gender:	0.83	
Pol Dynasty:	0.83	
High Office:	0.83	
Affiliation:	0.83	
Age:	0.83	
Total:		1.0

Results

4.3 Candidate Use of 'Change' Rhetoric

Figure 1 shows the results of the textual frequency for all presidential nominees in the last seven election cycles:

⁹ While this transformation is suitable for our study, it is far from ideal, as it can exceed the necessary bounds of [0,1] easily (for example, a 36-year old candidate would receive a score of 2.0). Several transformations (including exp-based) were considered, but the current form was kept for the useful way it spreads out the values in our dataset.

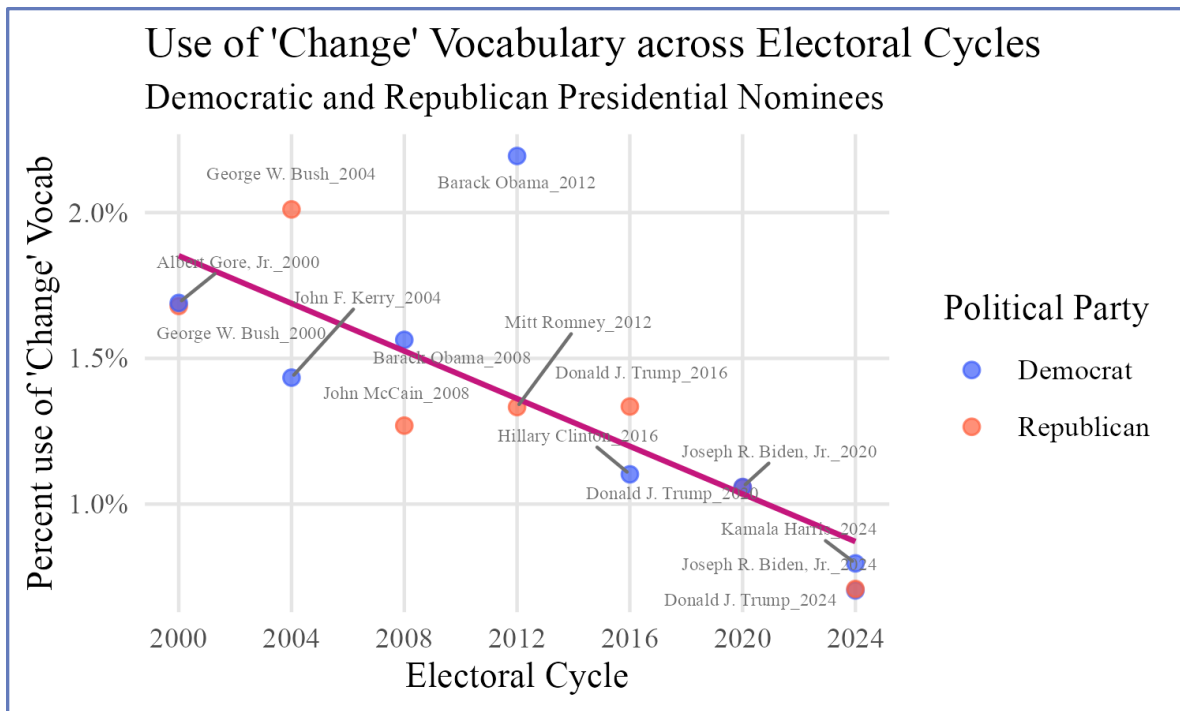


Figure 1, Usage of 'Change' Rhetoric in Campaign Documents for Presidential Nominees

The first thing we note is that there is no clear ownership over usage of 'change-based rhetoric' by either the Democratic or Republican parties, which lends important credence to a fundamental assumption of this paper: that candidates in either party can run on a message of 'change' regardless of their party's ideological association with 'liberal' or 'conservative' values or policies. Most striking, however, is the clear downward trend we see across electoral cycles, which seems to indicate a significant decline in the use of verbal rhetoric of change in the last seven electoral cycles. This trend is explored in the following section. We also see that these results loosely coincide with the winning candidates in each election, with only 2000 (a famously contested electoral result) and 2024 (the only election with nominee-replacement in modern history) giving the opposite results.

4.4 Candidate Change Index (CCI) Scores

Figure 2 shows the result of the CCI exercise, which joins together candidate 'Posturing' and 'Perception'.

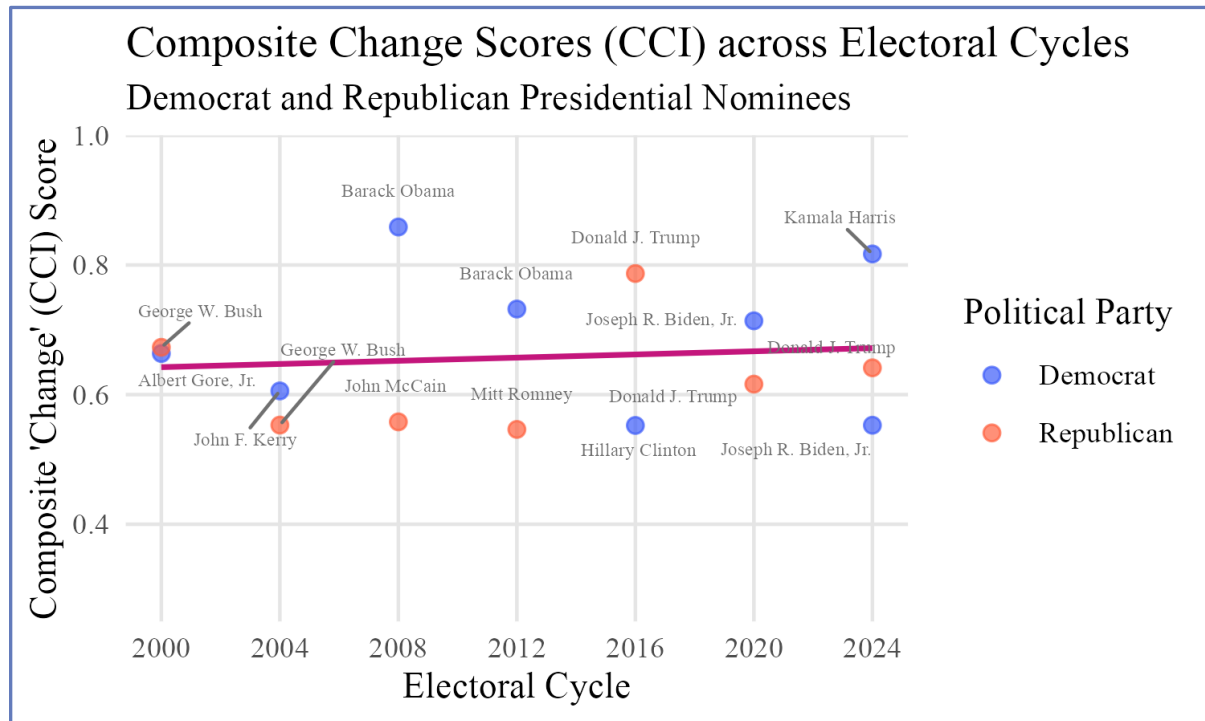


Figure 2, Results of Candidate Change Index Scoring, by Election

In comparison to the results of the previous section, which looked at only rhetoric, we see now an increase trend in the *overall* 'change' embodied by candidates over the last twenty-five years. This relative increase (relative to the severe decline seen in rhetoric) can be partially explained by the widening in the pool of candidates that now run for office. Beginning with Obama's 2008 candidacy, candidates belonging to minority groups such as ethnic minorities, women, and true political outsiders have begun to enter the presidential stage. These candidates, who are a departure (visually and substantively) from the traditional political establishment, perhaps embody change so evidently that they need to rely less heavily on rhetoric to communicate their 'change'.

We also see a new ranking of candidates in certain years. In 2000, Bush now has the higher score, while in 2004 Kerry takes this spot from Bush. Both of these results are likely due to the

inclusion of our 'Affiliation' variable, which is functionally a measure of anti-incumbency (as Clinton preceded Bush in the 90s and Bush went on to become the incumbent in 2004). Interestingly, we see some of the greatest gaps amongst candidates in the 2024 election, which is in line Biden and Trump's attributes as previous Presidents that were senior, white, and male.

Study Two: Identifying Change Voters and Theory Building

Methods

5.1 Vote Pattern Filtering

Following the results of the first study, we make use of the CCI and its 'change' ranking of candidates to identify voters who follow a pattern of voting for change. To do this, we filter respondents in the 2020¹⁰ ANES survey based on whether they voted in a pattern consistent with the results of each year's CCI rankings. In other words, we take respondents who report having voter in the following manner (consistent with the CCI change candidates for each year) in the 2012, 2016, and 2020 elections:

[2012: Obama] , [2016: Trump] , [2020: Biden]

The voters who follow this exact pattern are henceforth referred to as CVoters (for their pattern of 'Change Voting' across *all three* electoral cycles), while respondents who voted in 2020 but did *not* follow this exact pattern of voting are referred to as MVoters (given their 'Mixed' voting pattern). These CVoters and MVoters then become the basis for identifying the attributes that define a Change Voter.

¹⁰ We use the 2020 ANES survey because it is the only survey to ask respondents to recall their voting choices as far back as two elections prior.

5.2 Voter Attribute Independence Testing

In order to identify the unique attributes of CVoters, we take 61 questions¹¹ that represent a diversity of respondent beliefs, attitudes, and traits, pursuant to the literature and research goals of this paper, we apply a combination of three statistical tests to each question with the goal of determining whether the scores of CVoters differ *significantly*¹² from the scores of MVoters for that question. This is known in statistics as assessing whether two samples are independent of one another.

These statistical tests are the following:

1) Independent Samples T-Test

First, we apply the Independent Samples T-Test to all variables, given that it is valid for all of our data types (binary, ordinal, and semi-continuous), and that it is a convenient and simple blanket approach. However, we do so cautiously, aware that it can easily overlook more complicated distributional behaviour due to its assumption of normality in the data.

2) Wilcoxon Signed-Rank Test

The Wilcoxon test is used ordinal variables and assesses similarity of samples by looking observed counts in each of the possible ordinal values. Crucially, this makes the Wilcoxon test *non-parametric*, which makes it a nice complement to the T-Test, as it does not assume normality in the data. The Wilcoxon test is thus more resilient to distributions that may be skewed, multi-modal, uniform, or kurtosis-prone, as many of our variables are.

3) Fisher's Exact Test

Fisher's Exact is used for binary variables and is especially useful when there is a large imbalance of counts across groups (which is the case between our CVoters and MVoters). Each of these tests returns (among other measures) a p-value, which becomes central to our needs, as we consider only variables that receive p-values below 0.5 when formulating our theory of change voters later on.

¹¹ Found in the Appendices.

¹² $p < 0.05$

Results

5.3 Sample Size after Vote Pattern Filtering

From the 8,280 respondents in the 2020 ANES, 73% (or 6,029) indicated that they had voted for president in 2020, of which 55% voted for that year's Change Candidate, Joe Biden.

Of these, 86% voted in 2016, with an expectedly low¹³ number of those voters (7%) indicating that they voted for Donald Trump in 2016.

Finally, 83% of these indicated that had they also voted in the 2012 election, with 57% of them having voted for Barack Obama.

In total, this means that, from our original pool of 6,029 voters, 89 of them followed a voting pattern across all three elections consistent with our CCI rankings. While this subset initially seems worryingly small for deriving meaningful conclusions, the fact that we test on a large number of variables, use a variety of statistical tests, and pair this quantitative analysis an in-depth qualitative assessment of validity should relieve some concerns of a small sample size.

5.4 Attributes Statistically Associated with Change Voting

Our statistical independence testing results in twenty variables found to be distributionally independent¹⁴ between CVoters and MVoters. These variables, listed in descending order of p-value) are shown in Figure 3. We begin our analysis by grouping these variables into five broad themes and exploring basic directional tendencies for CVoters, before moving to full qualitative assessment and theory building.

¹³ Given that Joe Biden and Donald Trump belong to different political parties.

¹⁴ An interested reader may consult the Appendices, which graph each variables distribution beyond summary statistics.

Statistically Independent Attributes: C-Voters (2012, 2016, 2020 - ANES 2020)									
variable	description	scale	mean_MVot	mean_CVot	t_tstat	t_pval	w_pval	f_pval	f_oddsratio
V202158	Favourability Rating: Fauci	0-100 scale: 0 Terrible, 100 Great	69.26	84.20	6.23	0.00	0.00	NA	NA
V202173	Favourability Rating: Scientists	0-100 scale: 0 Terrible, 100 Great	80.09	86.57	4.53	0.00	0.01	NA	NA
V202332	Climate: impact level on United States	1. Not at all 2. A little 3. A moderate amount 4. A lot 5. A great deal	3.57	4.20	5.72	0.00	0.00	NA	NA
V202333	Climate: issue importance	1. Not at all important 2. A little important 3. Moderately important 4. Very important 5. Extremely important	3.35	3.83	4.34	0.00	0.00	NA	NA
V202439	Left-Right self-placement	0. Left <-> 10. Right	5.63	4.93	-4.20	0.00	0.01	NA	NA
V202431	Matters who is in power	5-point scale, 1. It doesn't make any difference 5. It makes a big difference	4.25	3.81	-3.26	0.00	0.00	NA	NA
V202264	As world changes, morals should change	1. Agree strongly, 2. Agree somewhat, 3. Neither agree nor disagree, 4. Disagree somewhat, 5. Disagree strongly	3.02	2.68	-2.88	0.01	0.03	NA	NA
V201367	Importance: Branches of Govt check one another	1. Not important at all 2. A little important 3. Moderately important 4. Very important 5. Extremely important	4.45	4.16	-2.86	0.01	0.00	NA	NA
V202427	Government Performance - 4yrs	1. Very good job, 2. Good job, 3. Bad job, 4. Very bad job	2.82	3.06	2.83	0.01	0.05	NA	NA
V202216	Differences in Major Parties	1. Yes, differences, 2. No, no differences	1.06	1.16	2.55	0.01	0.00	0.00	0.33
V201115	How Hopeful about how things are going in US	1. Not at all 2. A little 3. Somewhat 4. Very 5. Extremely	2.50	2.22	-2.52	0.01	0.04	NA	NA
V202430	State of Economy - 12months	1. Gotten much better, 2. Gotten somewhat better, 3. Stayed about the same, 4. Gotten somewhat worse, 5. Gotten much worse	3.66	3.93	2.46	0.02	0.19	NA	NA
V201400x	Income Gap - Today vs 20yrs ago	1. Much larger 2. Somewhat larger 3. About the same 4. Somewhat smaller 5. Much smaller	1.70	1.48	-2.45	0.02	0.05	NA	NA
V201368	Importance: Elected Officials consequences if misconduct	1. Not important at all 2. A little important 3. Moderately important 4. Very important 5. Extremely important	4.64	4.44	-2.26	0.03	0.01	NA	NA
V201119	How Happy about how things are going in US	1. Not at all 2. A little 3. Somewhat 4. Very 5. Extremely	1.92	1.72	-2.22	0.03	0.14	NA	NA
V202468x	Family income	1. <-> 22.: 1. Under \$9,999, 22. \$250,000+	12.41	10.95	-2.04	0.04	0.05	NA	NA
V201225x	Is Voting a Duty or a Choice	1. Very strongly a duty 2. Moderately strongly a duty 3. A little strongly a duty 4. Neither a duty nor a choice 5. A little strongly a choice 6. Moderately strongly a choice 7. Very strongly a choice	3.01	3.56	2.03	0.05	0.04	NA	NA
V202162	Favourability Rating: Labor Unions	0-100 scale: 0 Terrible, 100 Great	58.06	62.82	1.94	0.06	0.05	NA	NA
V202138y	Recall: VP Pence	0. Incorrect, 1. Correct	0.95	0.90	-1.71	0.09	0.01	0.04	2.33
V201366	Importance: News Orgs. be free to criticize	1. Not important at all 2. A little important 3. Moderately important 4. Very important 5. Extremely important	3.80	3.57	-1.69	0.10	0.04	NA	NA

Figure 3, CVoter Attributes with Statistical Independence, Relative to MVoters.

Theme 1: Economic Situation/Economic Perspective

Variables: [12], [13], [16], [18]

*Conclusion: CVoters have **Worse** Economic Situation/ Perspective*

We see first that CVoters score lower on questions of both *personal* economic situation and economic *outlooks* when compared to MVoters. Family Income for CVoters on average is \$7,000 USD lower¹⁵, while their View of the Economy in the Last 12 Months is also worse, and their perception of the Income Gap in the Last 20 Years is that the gap has grown. Finally, they tend to have a more favourable view of Labour Unions, which is discussed in the qualitative assessment section.

Theme 2: Belief in Science and Progress

Variables: [1],[2], [4], [7]

*Conclusion: CVoters have a **Positive** view of Science and Progress*

We see a strikingly high level of favourability for 'Scientists' and for Dr. Anthony Fauci¹⁶ (nearly a 15-point difference compared with MVoters). Further, we see a greater belief in the importance of the issue of Climate Change, as well as an 'open-mindedness' in their view of how 'Morals should Change with a Changing World'.

Theme 3: View of Future and Past

Variables: [3],[8], [11], [15]

*Conclusion: CVoters have a **Negative** view of Past and Future*

First, they score lower on rating 'Government Performance in Last 4 Years', which shows a negative view of the past. In the sentiment questions asking how 'Hopeful' and 'Happy' the respondent feels about 'How Things are Going in the US', CVoters score lower on both measures, indicating a discontentment with the present *and* prospects for the future. Finally, they

¹⁵ Coded average values of ~12.4 for MVoters and ~11.0 for CVoters map to the ANES income scale as 11: \$60,000-64,999, 12: \$65,000-69,999

¹⁶ Chief Medical Advisor to the President at the time of this survey.

tend to rate the future Impact of the 'Climate on the US' as having more impact, solidifying a pessimistic view of the future.

Theme 4: Partisan Alignment

Variables: [5],[6], [10], [19]

*Conclusion: CVoters feel **Non-Alignment** with Established Parties*

First, we see a strong tendency towards the center when asked to 'Self-Place on a Left-Right political Scale'. However, a simple mean score of 4.93 for this variable may hide more interesting truths, which are further explored in the following section. Next, they tend to not 'Believe it Matters Who is in Power', and see less 'Difference in the Major Parties'. Importantly, their slightly lower recall of Pence's position (90% vs. 95%) hides an interesting fact that is explored in further discussion.

Theme 5: Safeguarding Democracy

Variables: [9],[14], [17], [20]

*Conclusion: CVoters feel **Relative Apathy** towards Democracy*

These voters tend to place *less importance* on issues relating to the healthy functioning and preservation of a strong democracy. They rank lower on the perceived importance of having 'Branches of Government Check One Another', that 'News Organizations be Free to Criticize', and that 'Elected Officials Receive Severe Consequences for Misconduct'. Finally, they see 'Voting as less of a Duty' compared to their MVoter counterparts.

Discussion: Towards a Theory of Change Voters

6.1 Towards a Theory of Change Voters

Following the results of our quantitative studies up to this point, we use their insights as a starting place and refine these ideas using qualitative assessment to arrive at a unified Theory of Change Voters.

We begin with the theme of Economic Situation/Perspective and highlight the fact that the results of our independence testing have very clearly validated a central aspect of our hypothesis: existing in a state of constant financial distress. This is substantiated by the fact that we saw not only *one* measure of economic distress in the results, but a wide variety of indicators that together point to a very complete definition of financial distress. We see that these voters had *not only* lower incomes on average, but also a perception of the economy as getting worse and an awareness of the income gap widening, which is a correlate to the stagnation of real wages¹⁷. Additionally, the timeline captured by the Income Gap variable (20 years) is further association with our own understanding of real wage stagnation since the early 2000s. Finally, we remember the high favourability ratings given to labour unions, and take this (in conjunction with the observed lower average household income levels for CVoters) as signaling that CVoters may be more likely to belong to blue collar industries, where labour unions are more commonplace (Gelman et al, 2010), and where the evaporation of jobs and economic prosperity has been felt the most strongly in the American 21st century (Gelman et al, 2010).¹⁸ All of these taken together create solid evidence for including *persistent economic distress*¹⁹ as a central pillar of our Theory of Change Voters going forward.

We move to the second important aspect, which was an unspoken assumption of the paper (unspoken in that it is an assumption in swing-voting more broadly): that of a roughly 'moderate' political leaning of voters. Here, however, we find that our results are far more interesting than a simple 'centrist' explanation. Figure 4, showing respondent scores for the ANES variable 'Left-Right Self-Placement', evidences this exactly.

¹⁷ As industry becomes increasingly extractive, inflationary, and price-gouging, the profits taken home by the ultra wealthy increase at a rate higher than that of the median income.

¹⁸ This would also explain their voting for Joe Biden (a famously blue-collar and labour union candidate) in 2020, and perhaps prime us to believe that they would have voted for the manufacturing candidate, Donald Trump, in 2024.

¹⁹ On this idea, the variables identified in section 5.4 as belonging to the theme of Perceptions of Future and Past correlate heavily with (and are arguably founded in) economic distress. The observed negative outlook that CVoters hold on the future (low Hope) and low satisfaction with the present (low Happiness) further validate the theory of a strong desire for change. For the purposes of our study, we wrap these results into the broader idea of economic distress as, in our theory, economic distress is an inceptive source of distress and is arguably a confounding variable between voter 'Outlook' and 'Wanting Change'.

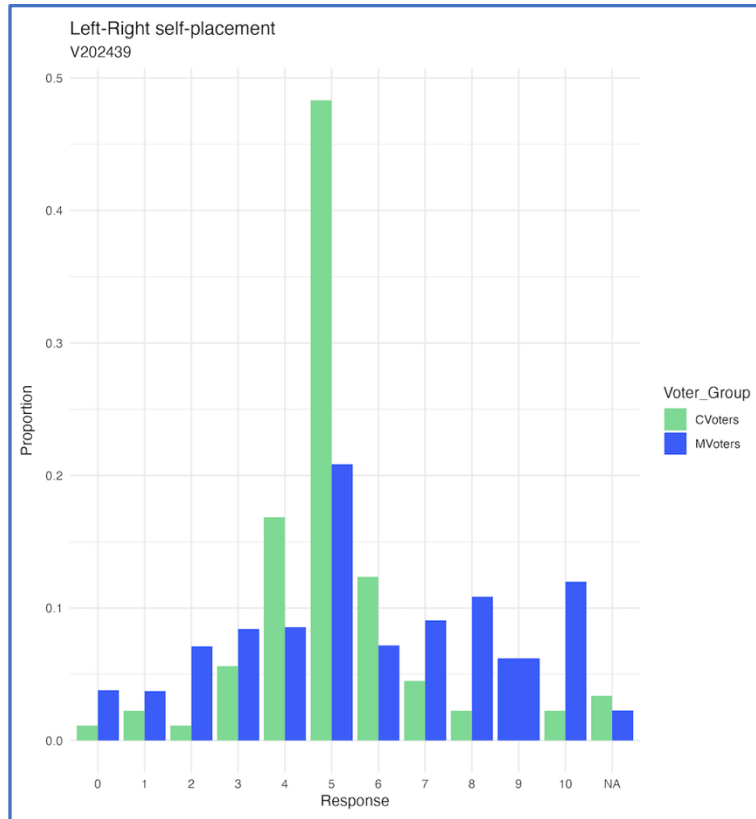


Figure 4, Distributions for CVoters and MVoters in 2020 Variable 'Left-Right'

We see that CVoters are not only grouped around the center of the scale (and thus begetting an initial explanation of 'centrism'), but also that they are in fact *strikingly unimodal* in their grouping around the specific response value of '5' (mapping to neither-left-nor-right). The sheer frequency of this value amongst CVoters, especially when compared to the less-modal distribution for MVoters, makes us understand this behaviour as not a *centrist leaning*, but instead as a *rejection* of the given political spectrum and a conscious non-alignment with the ideological values available in America's de-facto two-party system.²⁰ This formulation of Change Voter political non-alignment is also supported by the observed attitudes of apathy towards the healthy functioning and safeguarding of Democracy in variables such as passivity towards the balance of power between the branches of government.²¹ The observed behaviours

²⁰ The proportion of votes for independents is briefly studied, attempting to describe the discrete probability of voting for candidates across different elections. Unfortunately, this fascinating question is outside the scope of this paper.

²¹ This apathy towards protecting Democracy is consistent with Harris's failure in 2024 to convince the electorate of Trump's existential threat to Democracy, arguably contributing to her loss in the contest.

point us then towards the second key pillar of our Theory of Change Voters, which is a *conscious non-alignment and rejection of the political environment*.

Finally, we lean on the results (and non-results) of the Office Recall and Science questions to formulate our final pillar.

First, we speak on the fact that, of the four 'Office Recall' questions (Pence, Pelosi, Merkel, Roberts²²) included in the original 61 variables passed through the independence testing of this study, only one (Pence) gave a significant result in the direction of our original hypothesis: that Change Voters would be lower-information individuals.

This is a fundamental difference of what was expected in the results, and so forces us to reconsider the presence of information and political knowledge in the theorized Change Voter. Supplementing this new view of Change Voters, we take the high favourability ratings for 'Scientists', belief in Climate Change, and positive view of Dr. Anthony Fauci²³ as further evidence in the direction of these voters being higher-information (or at least not 'low-information) individuals.

Observing our Economic, Ideological, and Informational pillars, we can thus formulate a single, unified 'Theory of Change Voters', which describes Persistent Economic Distress, Conscious Non-Alignment, and Sufficient Information Levels.

²² Chief Justice of the SCOTUS

²³ Though these belief in science indicators do help us cement the informational aspect of voters, we do not include scientific themes in our unified theory as a standalone pillar, as 'Scientists' played an outsized role in the 2020 election (given the Covid-19 Pandemic) and a belief in Climate Change is a partisan issue.

Study Three: Change Voter Theory Validity Testing

Methods

Following our quantitative and qualitative formulation of a Voter Change Theory, we apply it to all ANES datasets (2000-2020²⁴) to assess the validity of the theory. In doing this, two methods were devised:

- 1) A discrete filtering of voters to arrive at the probability of choosing the candidate of change
- 2) A continuous scoring of voters to measure their affinity for change and their propensity for choosing the change candidate

Unfortunately, the first method is discarded because of how dramatically the filtering reduces dataset sizes (beyond the reach of analysis). We move forward then with the continuous measure.

7.1 Continuous Measure, Voter Propensity for 'Change Voting'

This approach forgoes any discrete filtering and instead gives each respondent a 'score' representing propensity of voting for change, from which we can derive overall insights.

This 'Change Voter Score' is composed of three variables, representing the three pillars of our Theory of Change Voters. These variables were selected based on how closely they reflect the theorized aspect that the pillar is meant to capture. They are as follows:

Pillar: Economic Distress

Selected Variable: Income Gap

Other variables were considered, including Household Income and State of Economy in Last 12 Months. However, Income Gap Change in Last 20 Years is selected because it 1) closely reflects a *worsening* economic anxiety, 2) is agnostic of income level²⁵ and 3) has a broad scope, not solely focused on the past year.

²⁴ While 2020 served as the basis for identifying CVoter attributes, we see no issue in using it for validity testing, given that we conduct a fundamentally different measure in this study as opposed to the pattern-based filtering of section 5.1.

²⁵ Income level has been shown to be highly correlated to party lines (Dem issue until 2012, Rep from 2016 onwards) (Gelman et al, 2010),

For the year 2000, no question on the income gap question was asked. In its stead, we "US Economy In Past Year" for 2000.

Pillar: Partisan Non-Alignment

Selected Variable: Left-Right Self

Other variables considered included Liberal-Conservative Self-Placement, Trust in Washington D.C., and Feelings Towards Republican or Democratic Parties. However, given the strong result for the variable Left-Right Self-Placement discussed in section 6.1, we select it for our validation testing.

For the year 2000, this question was not asked. In its stead, the question "Self Placement Lib-Con Scale" is used.

Pillar: Information Level

Selected Variable: Office Recall

Other considered for this pillar were Attention to Politics and Frequency of Listening to Political Radio Talk Shows, but they were dismissed for the fact that 'attention to politics' does not equal a respondent's information level (only the amount of attention devoted to the political sphere), and that the medium of Radio/ Television as sources of news has diminished greatly since 2000, respectively. Thus, we take Office Recall as our selected variable for this pillar, and the specific question asking respondents in each survey to name the position of the current Speaker of the House.

For the year 2008 all responses to Office Recall questions are restricted to the viewer, meaning that they are not usable. In place of this, we take "Attention to Politics and Elections" for 2008. For the year 2000, an Office Recall Question for Speaker of the House was not asked. In its stead, the question Office Recall Prime Minister of Britain Tony Blair is used.

Respondent values for each of these variables are then 'scored' depending on the available categories in each variable, similar to the CCI developed for candidates. In terms of scoring direction, voters who note a larger income gap receive higher ECON scores, higher LEFTRIGHT scores the closer to the center the respondent is (with extreme left or extreme right voters receiving gradually lower scores, in a 'folded scale'), and higher INFORMED scores to voters that can accurately identify the office (position) of the politician in the survey question.

These three variable scores are then weighted according to the below scheme:

$$\text{Change Voter Score} = (0.5 * \text{ECON}) + (0.35 * \text{LEFTRIGHT}) + (0.15 * \text{INFORMED})$$

This weighting scheme²⁶ is based on two things.

1) ECON is given the highest weight, as the central tenet of this paper is that the median voter is in a state of constant economic anxiety, and that this is the inciting phenomenon that begins to form a change voter.

2) INFORMED is given the lowest weighting due to the fact that it comes largely from our qualitative theory formulation and had little statistical support in the direction we are employing it now.²⁷

²⁶ Many weighting schemes were tested. Beyond the qualitative reasons for choosing this scheme, it is also seen as a valid formulation due its stability. Changes smaller than 5 points to the weighting scheme for any variables do not alter the results.

²⁷ Readers will remember that we saw a rejection of CVoters being *low* information, but no evidence that they would be *high* information individuals.

Results

7.2 Mapping Voter Change Scores to Candidate Change (CCI) Scores

In assessing these results, we begin first by remembering that our 'Voter Change Score' can be interpreted as the propensity of a voter to vote for the Change Candidate in the given election year. Figure 5 show this relationship, plotting Voter Change Scores frequencies²⁸ against the CCI scores of the candidates for each year, coloured by party affiliation. Importantly, this means that the candidate at the 'top' of the y-axis is always the one with a higher CCI, and thus the Change Candidate by our measures.

Additionally, we see two other elements.

First, we see a horizontal LOESS (Locally Estimated Scatterplot Smoothing) curve, which helps us visualize the frequency of Voter Change Scores, as the LOESS curves gravitate (fit) towards greater concentration of values. We can interpret it as a sort of reverse density function, where a lower density of points is reflected with a 'smaller' function.

Finally, we see diagonal chord lines in light purple, which point us to the mean Voter Change Score for each of the two candidates in an election. The 'X' at one end of each chord denotes which of the two candidates had voters with greater Voter Change Scores, on average. In other words, the 'X' tells us which candidate's voters were more 'Change'-minded.

²⁸ Importantly: Voter Change Scores exist on non-continuous scale, given that the questions (variables) behind them are ordinal or binary in nature, resulting in a smaller number of possible values.

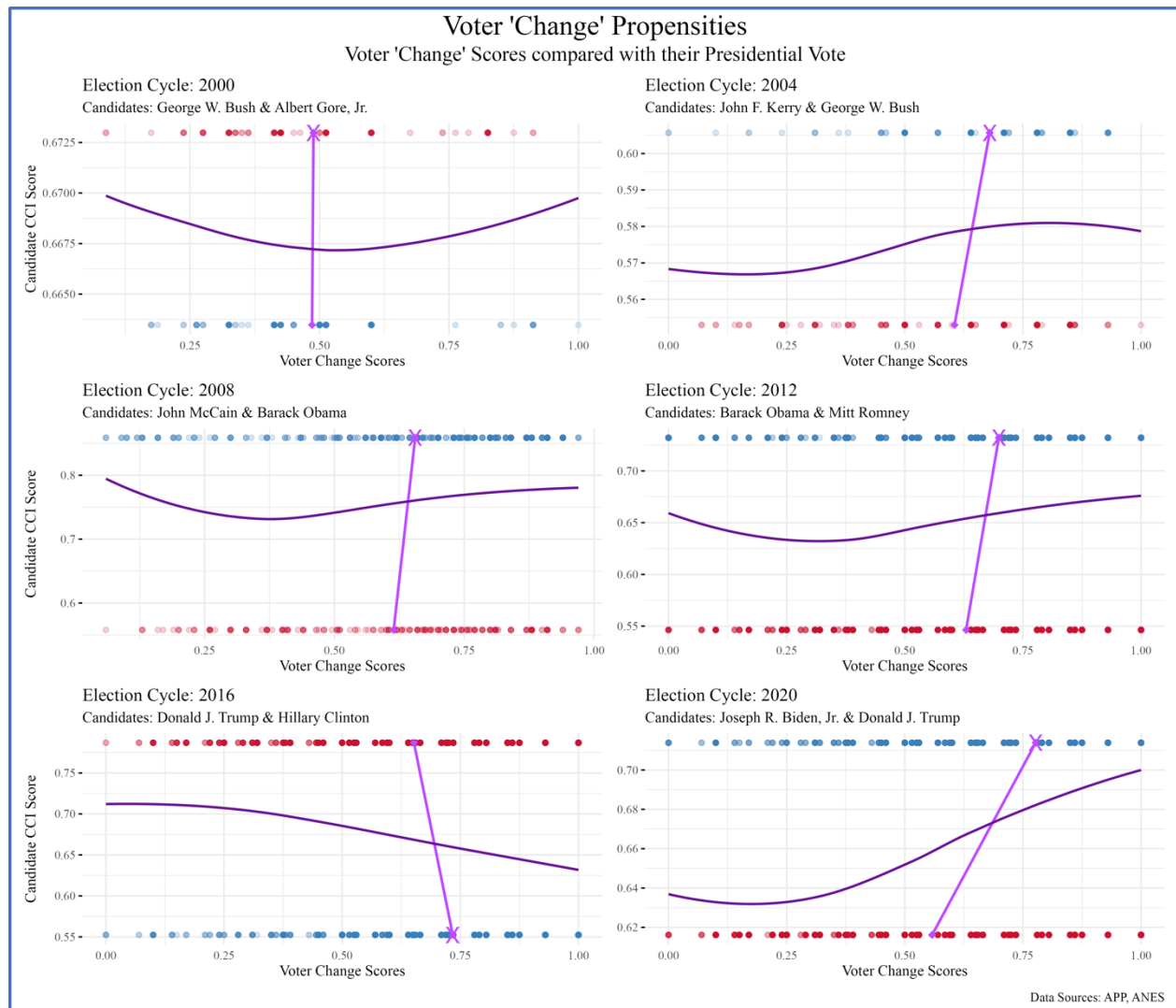


Figure 5, Mapping of Voter Change Score Results to Candidate Rankings

First, we begin with the vital question: does our Theory of Change Voters let us predict the likelihood of voters to pick the Change Candidate in an election? The answer is 'maybe'.

We see that, for all elections except for 2016, the voters with higher Voter Change Scores were more likely to pick the Change Candidate, giving us a tentative validation rate of five out of six elections. However, there are more interesting trends to uncover here.

We turn to the shapes of the horizontal LOESS curves, which indicate the ends of the voter distributions which voter for each candidate.

In 2000, we see a very parabolic LOESS curve, meaning that Bush captured both the voters with lowest *and* highest change scores. We can dissect this to mean that voters with *fewer economic worries* and perhaps²⁹ *greater conservative leaning*, on top of the extremely change-prone voters towards the right of the scale.

In 2004, we see an 'S'-shaped (almost logistic) shaped LOESS curve, meaning that, as voters become more change-minded in 2004, they very steeply became more likely to vote for Kerry.

Obama has a very interesting pattern here, in that his LOESS curves for 2008 and 2012 are exceedingly similar in shape, and differ effectively only in 'intercept'. Their shape tells us that the *make-up* of his base was similar in both elections (capturing both low-change and high-change voters, but losing out on medium-low-change voters), but the 'lower intercept' of 2012 illustrates the fact that his win-margin dropped from 7% against McCain in 2008 to 4% against Romney in 2012. Crucially, this is consistent with the fact that the gap in CCI³⁰ between Obama and McCain was much larger than the gap between Obama and Romney, heavily validating both our measure of CCI and our Change Voter Theory testing.

In 2016 we see the only result inconsistent with the expected voting pattern, showing that, voters with higher propensity for change actually had a greater probability of voting for Clinton, who was decisively *not* the change candidate in 2016 (by our measure³¹). Explanations for this are discussed in the following section.

Finally, in 2020 we see the most stark behaviour in the LOESS curve. Here, voters *overwhelmingly* gravitated towards Biden as their propensity for change increased, strongly validating our CCI predictions.

²⁹ Remembering that our Left-Right measure was 'wrapped' so that extremists of both the Right and Left received a score close to 0, meaning that theoretically it is exactly as-likely that these would be extreme-left voters instead of conservatives.

³⁰ Figure 2 in Section 4.4

³¹ Her husband had been president for eight years between 1992 and 2000, she had worked as Secretary of State under Obama, and was a senior.

General Discussion and Implications

8.1 General Discussion and Implications

The question that needs to be addressed, once more, is the extent of validity of our Theory of Change. To do this, we must speak of the strong rejection we saw of our theory for 2016 in the previous section.

We saw that change-propense voted strongly for Hillary Clinton as their Voter Change Scores rose. Two explanations for this may exist for this, both of which address the validity of our proposed framework and methods used throughout the three studies.

First, we explain this as an 'underrepresentation' of the truth. We recall that, leading up to 2016, polling showed that Clinton was set to win the presidency, and that the aftermath of the election showed that that polls and surveys had famously erred in their assumptions, and under-surveyed the demographics that ended up voting for Trump (Philpot, 2018). This was due to the fact Trump succeeded in forming a new coalition of voters, many of which had not participated in previous elections (Philpot, 2018). Logically, this means that an important portion of the people that voted for Trump in 2016 are simply not represented in the ANES survey of that year, which is further evidence by a mismatch in the real-world popular vote margin and the one found in the ANES 2016 dataset³². Taken together, this tells us that the *source* from which we have so far derived our voter data and theorization (ANES) may be far from correct, and more broadly, that theories formed on the basis of data are only ever as good as the data itself. Practically, this means that arriving at a more 'accurate' Theory of Change Voters (or any new theory of voters) may be a flawed exercise, as it attempts to describe something that is inherently latent.

Second, we explain this through the recognition that our CCI measure is imperfect³³, and is severely underestimating the appeal that Clinton actually had to change voters by being the first-

³² Clinton received 48% of the popular vote in 2016 and Trump 46%, which is a 2% margin. Meanwhile, the ANES 2016 dataset paints Clinton as having a margin of 4%, indicating that the ANES of 2016 followed the same assumption errors that other polling/ surveying made, contributes to these bizarre 2016 Change Voter results.

³³ "All models are wrong, but some are useful."

ever woman to reach the presidential stage. Voters looking for change in 2016 may have seen Clinton's gender as *the* quintessential indicator of 'change' in a world ruled by men. Here we arrive at the fact that 'change' does not mean the same thing for everyone, and so once again, a unified Theory of Change Voters may be an inherently flawed exercise as many people may *consider* themselves 'change' voters, but be voting for very different *types* of 'change'.

LIMITATIONS and FUTURE WORK

9.1 Limitations

Several limitations exist, many of which have been discussed in earlier sections. These are summarized here.

First, if we believe our hypothesis, and change voting *is* becoming a greater it means that future elections elections hold more value for study, limiting the true testing of theory that we can undertake at the moment. For example, there is a good chance that 2028 will see the first presidential stage with an openly-gay candidate (Pete Buttigieg) or that it will be the second election in a row with a woman of colour as nominee (Alexandria Ocasio-Cortez).

Second, the Theory of Change Votes we develop is quite basic in that we only test using three variables. This is largely constrained by the available ANES data in two ways. First, the lack of consistency in survey questions available across ANES surveys makes an exact and consistent approach very difficult. Second, the ANES datasets are not large enough for our purposes, as was seen in each of our discrete filtering approaches. Applying a set of simple filtering rules results in sample sizes that arguably don't allow for valid generalization.

9.2 Future Work

Future Work could A) focus on different methods of assessing candidate rhetoric, B) examine the role of independents in presidential elections and their effects on the central candidates, and C) attempt a similar analysis using other data.

A) Future work could assess candidate rhetoric by going 'deep' as well as 'wide' (ie: analyze a small number of documents per candidate using more advanced methods such as TF-IDF in addition to a dictionary analysis spanning thousands of documents). Additionally, Word Embeddings could be used in place of a manual dictionary construction, leveraging advanced Data Science techniques over qualitative selection.

B) Future work could more closely examine the role of independents in the political system and the way they sell 'change' or take legitimacy away from the Democrat and Republican candidates attempting to do the same.

Finally, a future version of this paper and theory could draw from not only the ANES, but also the Cooperative Election Study, as the lack of demographic variables in the ANES severely limits the formation of theory.

Conclusions

10.1 Conclusions

While it is clear that the economic conditions that have formed over the last 25 years have had severe effects on the psyche of many Americans, and especially on those in disadvantaged positions, this paper's stance, positing that it has contributed to an increase in a kind of 'change' voting, has yet to be certainly observed.

Overall, our approach to formulating a Theory of Change Voters is strong, but flawed. We see that our theorization of the effects of real-wage stagnation on these kinds of voters is largely

correct, along with the fact that they tend to reject the left-right political paradigm. Crucially, however, we find that a portion of our theory, that these voters would be lower on the scale of information/ political knowledge, is disproven.

With this theory as a starting block, we can begin to understand that dissatisfied voters do care strongly about the state of the economy and the party of the current incumbent, but that these feelings hide also a larger swath of sometimes subconscious feelings that fit into a greater picture of contemporary society in an advanced, globalized democracy. There are winners and losers. There are insiders and outsiders. There are those who fight for democracy because it is 'self-evident' and those who are apathetic to its death. In all of this, the only constant is change.

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Appendices

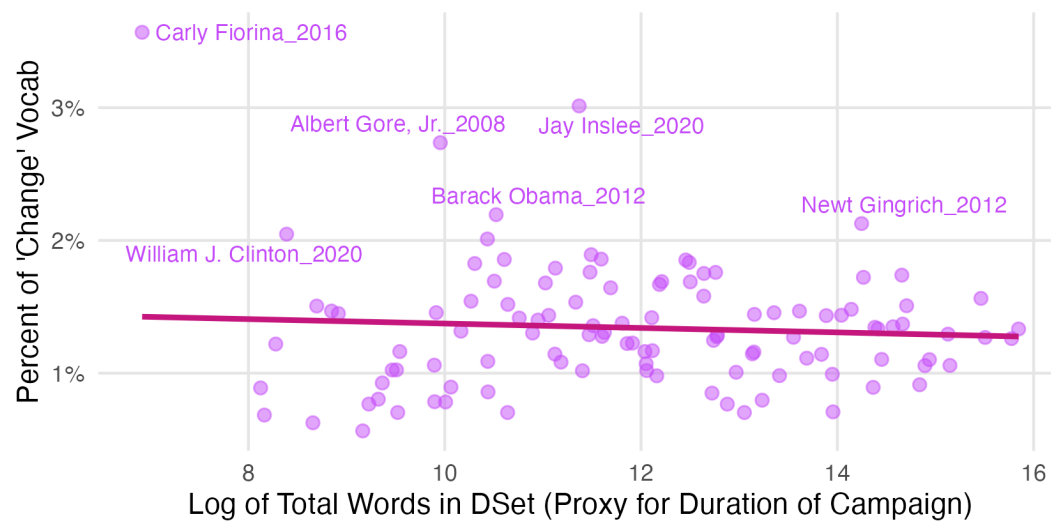
Change Dictionary

stem	length_ngram	category
change	unigram	direct_change
shift	unigram	direct_change
reform	unigram	direct_change
revolut	unigram	direct_change
transform	unigram	direct_change
overhaul	unigram	direct_change
reset	unigram	direct_change
pivot	unigram	direct_change
reimagin	unigram	direct_change
redo	unigram	direct_change
rebuild	unigram	direct_change
modify	unigram	direct_change
restructur	unigram	direct_change
innovat	unigram	direct_change
progress	unigram	direct_change
advanc	unigram	direct_change
forward	unigram	forward_looking
future	unigram	forward_looking
new	unigram	forward_looking
next	unigram	forward_looking
vision	unigram	forward_looking
potential	unigram	forward_looking

fresh	unigram	forward_looking
modern	unigram	forward_looking
alternativ	unigram	forward_looking
evolving	unigram	forward_looking
evolution	unigram	forward_looking
emerging	unigram	forward_looking
adapt	unigram	forward_looking
broken	unigram	contrast_statusquo
outdated	unigram	contrast_statusquo
fail	unigram	contrast_statusquo
old	unigram	contrast_statusquo
established	unigram	contrast_statusquo
establishment	unigram	contrast_statusquo
gridlock	unigram	contrast_statusquo
swamp	unigram	contrast_statusquo
corrupt	unigram	contrast_statusquo
elit	unigram	contrast_statusquo
stagnat	unigram	contrast_statusquo
obsolete	unigram	contrast_statusquo
rigid	unigram	contrast_statusquo
stuck	unigram	contrast_statusquo
backward	unigram	contrast_statusquo
challenge	unigram	action_oriented
break	unigram	action_oriented
disrupt	unigram	action_oriented
overhaul	unigram	action_oriented
reject	unigram	action_oriented
undo	unigram	action_oriented
replac	unigram	action_oriented
repeal	unigram	action_oriented
dismantl	unigram	action_oriented
rework	unigram	action_oriented
revamp	unigram	action_oriented
uproot	unigram	action_oriented
renovat	unigram	action_oriented

Relationship Between Duration in Race and Change Rhetoric

Duration in Race vs. Degree of 'Change' Vocabulary



Statistically Independent Variables for CVoters.



Statement of Authorship

I hereby confirm and certify that this master thesis is my own work. All ideas and language of others are acknowledged in the text. All references and verbatim extracts are properly quoted and all other sources of information are specifically and clearly designated.

In accordance with the Hertie School's AI Guidelines (February 2023), I declare that I have used AI tools, specifically Open AI's 'ChatGPT' family of Large Language Models (LLMs) and Undermind's LLM for assistance with coding and thesis preparation. These tools were used in the following manner:

- Code Generation and Debugging
- Developing the Structure of the Document
- Research Assistance for Literature Review Sections and Abstract

I confirm that all AI-generated content has been carefully reviewed, edited, and verified by me for accuracy, and that I take full responsibility for all content, including any factual errors or inaccuracies that may exist in the final document. The use of these tools has been in accordance with the Hertie School's Policy on Academic Integrity, with all significant contributions properly acknowledged.

I confirm that the digital copy of the master thesis that I submitted on the **28th April 2025** is identical to the printed version I submitted to the Examination Office on the **29th April 2025**.

DATE: 28th April 2025

NAME: Juan Pablo Brasdefer

SIGNATURE:

A handwritten signature in black ink, appearing to be 'JP Brasdefer', written over a horizontal line.